

Tissue Classification in MR Images Using Hierarchical Segmentation

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Abstract Simple threshold and gradient based segmentation algorithms are unable to provide reliable soft tissue classification in MR images. To obtain improved estimates, we have developed a new unsupervised hierarchical segmentation algorithm upon which our tissue classification scheme is based. The MR images are modeled as a first order Gauss-Markov random field (GMRF) with unknown parameters. Through testing hypotheses of homogeneity for either labeled or unlabeled data, the segmentation algorithm seeks to group the pixels into connected regions which are homogeneous GMRFs. This is achieved in three steps. The image is represented by a quadtree and a split-and-merge procedure is applied to find large homogeneous regions within the image. This is followed by a segmentation refinement step involving connected component labeling and region growing. Here, the use of a hierarchical approach allows the algorithm to learn about the image by first processing the easy-to-classify pixels while delaying decisions on the hard pixels until more information has been gathered. Finally, a novel graph theoretic clustering algorithm is developed to reduce the number of resulting connected regions. This clustering method is globally optimal for a properly defined family of cost functions.

1 Introduction

Two basic approaches to MR image segmentation have been described in the literature: i) methods based on edge detection; and ii) methods based on the statistics of regions corresponding to different tissue types. Ehrlicke [1] and Bomans et al [2] have developed MR segmentation algorithms based on the Marr-Hildreth edge detector. While this approach guarantees the formation of regions with closed boundaries, these boundaries are often inaccurate, particularly between regions with small contrast differences. Various attempts to overcome this problem are discussed in [1] and [2]. The second approach does not directly seek the boundary between regions, but rather groups pixels into regions based on some similarity measure. These methods may be either supervised [3], [4] or unsupervised [5], [6]. Since estimated MR parameters vary from machine to machine and patient to patient, an unsupervised methods appears more attractive; it is this approach that we adopt in this paper.

We describe a tissue classification scheme for magnetic resonance (MR) images of the human brain, which falls into the second category. Our classification scheme is based on an un-

supervised hierarchical segmentation technique. The images under study are modeled as a mosaic of regions where the pixel intensities in each region are modeled by a first order homogeneous Gauss-Markov random field (GMRF) with unknown parameters [7]. The choice of a first order neighborhood for the system was motivated by the fact that the variations within a region of an MR image corresponding to a single tissue type generally exhibit very local correlation rather than large scale textual characteristics, and consequently this low order model should be appropriate.

Image segmentation based on Markov random field models has been widely studied. Among the numerous papers in the literature, there are two general approaches which are closely related to this work: Bayesian [8], [9] and hierarchical "maximum likelihood" [10], [6]. This paper is focused on developing an unsupervised hierarchical segmentation scheme using homogeneity testing for GMRFs and a new region clustering technique. The unsupervised approach is preferable to a supervised method in situations such as tissue classification of MR images where the parameters for a given tissue may vary from patient to patient and from scanner to scanner. The goal of this segmentation algorithm is to group the pixels into regions which, under a suitable hypothesis, are homogeneous GMRFs. A brief discussion of homogeneity testing for either labeled or unlabeled data will be presented and different tests for homogeneity will be suggested.

The segmentation is implemented in three steps. A split-and-merge procedure is applied to find large homogeneous regions within the image. A segmentation refinement step then follows which involves connected component labeling and region growing. Here the use of a hierarchical approach allows the algorithm to learn about the image by first processing the easy-to-classify pixels while delaying decisions on the hard pixels until more information has been gathered. Finally, a novel graph theoretic clustering algorithm is developed to reduce the number of resulting connected regions. The algorithm uses an undirected graph to represent the image with each vertex corresponding to a connected region. This clustering method is globally optimal for a properly defined family of cost functions.

2 Image Modeling

Let X denotes a real valued image defined on the image domain Z , usually a rectangular lattice. Let Z be composed by K exhaustive and mutually exclusive subsets $\{Z_k\}_{k=1}^K$, and X_k , the subimage defined on Z_k , be modeled as a Gauss-Markov

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random field (GMRF). The image X can then be described by its joint probability density function (pdf) given by :

$$f(X) = \prod_{k=1}^K f(X_k),$$

provided the X_k and X_j are independent for $k \neq j$. Based on this composite image model, the segmentation becomes a problem of fitting GMRF parameters and estimating the partition $\{Z_k\}$.

The GMRF model for X_k on Z_k is defined by the following joint probability density function [7] :

$$f(X_k) = \frac{|B_k|^{\frac{1}{2}}}{(2\pi\sigma_k^2)^{\frac{N_k}{2}}} e^{-\frac{1}{2\sigma_k^2}(X_k - \mu_k \mathbf{1})^T B_k (X_k - \mu_k \mathbf{1})} \quad (1)$$

where N_k = number of pixels on Z_k , $\mathbf{1}$ denotes an $N_k \times 1$ vector consisting of all 1's, and for a first order neighborhood

$$B_k(i, j) = \begin{cases} 1 & \text{if } j = i \\ -\beta_{1k} & \text{if } j \in N_H(i) \\ -\beta_{2k} & \text{if } j \in N_V(i) \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where $N_H(i)$ and $N_V(i)$ denote the sets of horizontal and vertical nearest neighbors of pixel i , respectively. Here β_{1k} and β_{2k} should be properly chosen so that B_k is positive definite. In our study only a first order neighborhood and constant mean are considered. However we note that extensions to parameterized mean and higher order neighborhood are possible.

Given an observation X_k on Z_k , evaluating the likelihood $f(X_k)$ for fixed parameter set $\theta = (\mu, \sigma^2, \beta_1, \beta_2)$, and finding the maximum likelihood estimate of θ are both costly operations, which becomes prohibitive when Z_k is a large irregularly shaped region as often encountered in segmentation. A popular choice of approximation is the use of the pseudo-likelihood [11]. A more accurate approximation was proposed in our earlier papers [12], [6].

3 Homogeneity Testing

In a previous paper [6], we presented an analysis on the statistical testing of GMRFs for homogeneity. The relevant results will be summarized here without detailed justification.

Let X be a subimage defined on an arbitrary region Z consisting of N pixels. We want to test whether X is a realization of a homogeneous GMRF (the null hypothesis). Under the alternative hypothesis H_1 , X is a mosaic of homogeneous GMRFs, i.e. $X = (X_1, X_2, \dots, X_K)^T$ where K is the number of GMRFs in X and X_i denotes the N_i samples from the i -th GMRF parameterized by $\theta_i = (\mu_i, \sigma_i^2, \beta_{1i}, \beta_{2i})$. The null hypothesis H_0 may be more conveniently stated as $\theta_i = \theta, \forall i$ for X partitioned into K groups. Here only tests against the class of equal and known variance alternatives, i.e. $\sigma_i^2 = \sigma^2, \forall i$, are studied. This class excludes images in which the variance is the only distinguishable feature.

When the data are labeled, i.e. the partition $\{X_i\}$ of region X is given, testing homogeneity falls within the conventional framework of statistical hypotheses testing [13]. For the limited class of alternative hypotheses stated above, the likelihood ratio test is used, which rejects the null hypothesis H_0 if

$$\tilde{W}_K = \frac{1}{\sigma^2} (N\sigma^2 - \sum_{i=1}^K N_i \sigma_i^2) + \ln |B^*| - \ln |\hat{B}| > C, \quad (3)$$

where $\hat{\mu}$ and $\hat{B}$ denote the MLE of μ and B under H_0 , and μ_i^* and B_i^* denote the MLE of μ_i and B_i under H_1 for the i -th subimage. The MLE of B refers to the matrix constructed using the MLE of β_1 and β_2 . $\tilde{W}_K$ is approximately χ_{K-1}^2 distributed under H_0 .

Unfortunately X is often unlabeled which makes homogeneity testing more difficult. One common solution for testing homogeneity of unlabeled data is the following [14], [9] : divide X into K' subimages $\{X'_i\}$, usually of equal size, then test the homogeneity of X based on the statistics of $\{X'_i\}$. Examples of homogeneity criteria include $\tilde{W}_{K'}$ [9] and the maximum difference in the sample means of X and $\{X'_i\}$ [14]. This approach is simple but its power is usually poor due to the implicit assumption that the $\{X'_i\}$ are homogeneous themselves, regardless of the choice of homogeneity criterion; such an assumption is often invalid.

Two more powerful tests are proposed. The first is the so called dispersion test : reject H_0 if

$$\tilde{D}_N = \tilde{W}_N = \frac{N\sigma^2}{\sigma^2} > C. \quad (4)$$

A more elaborate test is also proposed : apply the dispersion test if the size of X is small; otherwise, divide X into K' groups and apply recursively the same test to all groups, then apply the $\tilde{W}_{K'}$ test defined in (3) only if all groups test homogeneous. This procedure is named as the hierarchical likelihood ratio test. This test has a clear purpose : to avoid testing homogeneity among subimages unless subimages themselves have a high probability of being homogeneous. The test can be naturally implemented within a hierarchical data structure, in our case a quadtree.

Unless all the GMRFs in the image partition are white, i.e. $\beta_{1k} = \beta_{2k} = 0$, the thresholds for the dispersion test and the hierarchical likelihood ratio test can only be set approximately for a given significance level.

4 Review of Network Flow

The problems in network flow theory which particularly interest us are finding the maximum flow from one vertex to another and finding the maximum flow between every pair of vertices. The methods for solving these problems provide the foundation of our graph theoretic algorithm for patch clustering which is described in section 5.

Let $\mathcal{G} = (\mathcal{V}, \mathcal{A})$ be a graph with vertices set $\mathcal{V} = \{v_1, v_2, \dots, v_M\}$ and arc set $\mathcal{A} = \{a_{ij}\}$ with a_{ij} denoting the arc between vertices v_i and v_j with capacity c_{ij} . A set of numbers f_{ij} defined on $a_{ij} \in \mathcal{A}$ are called flows from vertices s to t if they satisfy the following conditions:

$$\sum_{v_j \in \Gamma(v_i)} f_{ij} - \sum_{v_k \in \Gamma^{-1}(v_i)} f_{ki} = \begin{cases} F_{st} & \text{if } v_i = s \\ -F_{st} & \text{if } v_i = t \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

and $f_{ij} \leq c_{ij}$ for all $a_{ij} \in \mathcal{A}$. In (5) F_{st} denotes the flow from s to t , $\Gamma(v_i) = \{v_j | a_{ij} \in \mathcal{A}\}$ and $\Gamma^{-1}(v_i) = \{v_k | a_{ki} \in \mathcal{A}\}$ respectively.

Theorem 1 (Ford-Fulkerson theorem [15]) *The value of the maximum flow from a vertex s to vertex t is equal to the value of the minimum cut-set $(\mathcal{V}_m \rightarrow \bar{\mathcal{V}}_m)$.*

A cut-set $(V_o \rightarrow \hat{V}_o)$ separates s from t if $s \in V_o$, $t \in \hat{V}_o$ and $V_o \cup \hat{V}_o = V$. The value of $(V_o \rightarrow \hat{V}_o)$ is defined as:

$$f(V_o \rightarrow \hat{V}_o) = \sum_{a_{ij} \in (V_o \rightarrow \hat{V}_o)} f_{ij}.$$

The minimum cut-set $(V_m \rightarrow \hat{V}_m)$ is then the cut-set with the smallest such value.

The proof of the theorem can be found in [15]. Ford and Fulkerson [15] have also developed an algorithm for solving the maximum flow problem through finding the minimum cut-set from s to t . Observe that the minimum cut-set optimally partitions V into 2 subsets given the "seed" vertices s and t .

For an undirected graph G , the problem of finding the maximum flow between every pair of vertices or the "multi-terminal maximum flow problem" can be efficiently solved using an algorithm due to Gomory and Hu [16]. A complete description of their algorithm can be found in [17] and will not be given here. However, the following properties of undirected graphs are fundamental to the use of the Gomory-Hu algorithm in our work.

Theorem 2 ([15], [17]) Let F_{ij} be the value of the maximum flow from vertex v_i to vertex v_j of an undirected graph G . The flows then satisfy the relation:

$$F_{ij} \geq \min[F_{ik}, F_{kj}], \forall i, j, \text{ and } k. \quad (6)$$

The implication of the theorem is that if $(V_m \rightarrow \hat{V}_m)$ is a minimum cut-set separating v_i from v_j , then any minimum cut-set separating any pair of vertices in V_m (or in $\hat{V}_m$) will not partition further $\hat{V}_m$ (or V_m). This is crucial to the following theorem, and hence the optimality of our clustering algorithm.

Theorem 3 ([15]) Let $G^* = (V, A^*)$ be a hypothetical complete graph with the capacity of arc $a_{ij}^* \in A^*$ equal to the maximum flow F_{ij} between the corresponding vertices v_i and v_j and T^* be an arbitrary spanning tree of G^* , then T^* is cut-equivalent and flow-equivalent to the original graph G .

Cut-equivalency (flow-equivalency) means that G and T^* are indistinguishable as far as the minimum cut-set (maximum flow) between vertices is concerned. Once T^* has been constructed, the maximum flow between any pair of vertices and its corresponding minimum cut-set can be easily computed. The Gomory-Hu algorithm is used to efficiently construct the equivalent tree T^* which involves the successive solution of precisely $M - 1$ maximum flow problem with M being the number of vertices in G .

5 Segmentation Algorithm

Our new hierarchical MR image segmentation algorithm is implemented in three steps:

- (1) split-and-merge learning,
- (2) patch forming,
- (3) patch clustering using a graph theoretic approach.

Emphasis here will be placed on the patch clustering step. The first two steps of the algorithm will only be described briefly. A more detailed description can be found in [6].

5.1 Split-and-merge learning

A split-and-merge procedure [14] is applied to find large homogeneous regions within the image. The image is represented by a quadtree with the top of the tree corresponding to the entire image, the next level to the four quadrants of the image and so on. Starting from the top of the tree, we test homogeneity for the region corresponding to each node. Here homogeneity testing for unlabeled data is needed. The node is labeled accordingly: if the region is homogeneous a label is attached to that node together with the estimated GMRF parameters and the branch is terminated. If the region is inhomogeneous the node remains unlabeled and the same node labeling strategy is applied to its four quadrants. Furthermore each newly labeled region is tested for possible merging with previously formed homogeneous regions using tests for labeled data. This allows larger regions to be formed gradually, so that the GMRF parameter estimates becomes more and more reliable. As the node size becomes small, the parameters estimated from large homogeneous regions can provide supervision for deciding whether to merge the node to a larger region. On completion, the split-and-merge produces an image partition.

5.2 Patch forming

The split-and-merge procedure generally produces a large number of regions within the image, the purpose of the patch forming and clustering steps is to methodically reduce this number of regions. Each segmented region consists of a set of pixels assigned to the same label and may have many isolated subregions or patches, i.e. collections of spatially connected pixels. Through connected component labeling [18], all the patches for each region are identified and labeled accordingly: a different label is assigned to each patch except for those with very few pixels (≤ 4 pixels in our experiment) which are left unlabeled. The resulting unlabeled patches are then classified by growing the labeled patches: among all pairs of unlabeled patch and its neighboring labeled patch, select the one with the smallest mean difference and expand the labeled patch by attaching the unlabeled one to it. This patch growing process is repeated until the smallest mean difference exceeds a given threshold or no unlabeled patch remains.

5.3 Patch clustering

As expected, the number of patches resulting from the previous step is usually too large. In order to reduce the number of resulting patches, a new graph theoretic approach has been proposed to cluster those patches. This clustering procedure is optimal for a properly defined family of cost functions.

Graph formation :

An undirected graph is formed from the resulting patches: each vertex of the graph corresponds to a connected region (patch) of the image, and an arc is placed between a pair of vertices if their corresponding patches are neighbors of each other. A flow capacity is then assigned to each arc. The capacities may be arbitrary non-negative symmetric functions. We choose these functions to reflect the probability of the patches (vertices) belonging to the same region or tissue type. In our study the

following particular arc capacity function was used :

$$c_{i,j} = k_{i,j} \cdot e^{-\left(\frac{\mu_i - \mu_j}{\sigma}\right)^2},$$

where $k_{i,j}$ = number of neighbor pixels between vertices v_i and v_j , μ_i and μ_j are the sample means of patches i and j , and σ is a control parameter.

Clustering rationale :

Once the patch graph has been formed, the problem of clustering can be equivalently stated in terms of the removal of arcs of the graph to form mutually exclusive subgraphs. If two subgraphs should be formed from two seed vertices s and t , the resulting minimum cut-set separating s and t would provide an optimal partition in the sense that the flow (value of cut-set) from one subgraph to the other is minimized. Recall that smaller flow implies less resemblance between features of the two subgraphs. When no seed vertices are available, the optimal two subgraph partition would result from the minimal cut-set among all minimum cut-sets separating any pair of vertices. This is the well known problem in graph theory of finding the maximum flow between every pair of vertices of a graph, which has been efficiently solved by Gomory and Hu [17] through the construction of a flow equivalent tree. Thus by using the Gomory-Hu algorithm we can perform an optimal partitioning of the graph into a set of subgraphs where for each subgraph, the nodes of that subgraph represent patches whose union forms the region of the image corresponding to a single tissue type.

Clustering algorithm :

Step 1: Form the graph from the resulting patches.

Step 2: Find a flow equivalent tree for the graph.

Step 3: Remove successively arcs of the equivalent tree in ascending order of the maximum flows between every pairs, so that the graph can be partitioned into any desirable number of subgraphs.

This clustering algorithm produces an optimal partition of the graph into a given number of subgraphs, in the sense that the maximum of the inter-subgraph flows is minimized among all possible partitions of the graph into the same number of subgraphs. In addition constraints may be imposed on the resulting subgraphs, e.g. the number of pixels contained must be greater than a given threshold. In this case an arc will be removed only if the resulting subgraphs still satisfy the constraints. A partition obtained in this case is optimal subject to the constraints.

After clustering, most of the resulting patches are large and distinct from their surrounding patches. Therefore when applied to the problem of tissue classification in MR images, the tissue labeling of the patches can be easily performed through an interactive patch editing. In the future, we plan to replace this interactive component with an automated procedure based on a stored brain atlas.

6 Experimental Results

The hierarchical segmentation scheme has been applied for tissue classification of MR images of the human brain. Fig.1 shows

a typical cross-sectional MR image of the brain for a patient with a large lesion on left side of the brain. Note that the skull and external tissue have been removed from the image. After patch clustering a total of 83 subgraphs (regions) were obtained. To obtain the final results, these regions were further classified interactively into one of four types. Figures 2a-2d show the segmented regions corresponding respectively to (a) grey matter; (b) white matter; (c) ventricle; and (d) tumor. In each figure only pixels classified to that label are displayed while the rest are set to black. The union of the four regions is shown in Figure 3. We are currently working on the extension of this segmentation algorithm to tissue classification for 3-D MR images of the human brain.

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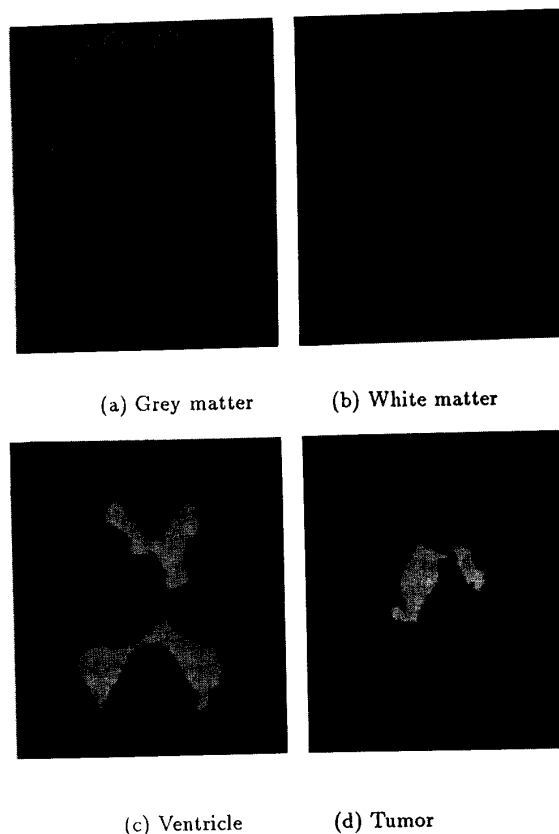


Figure 2: Resulting segmented regions

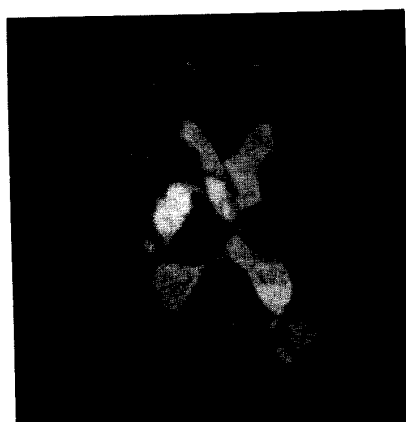


Figure 1: A cross-section MR image of brain with tumor

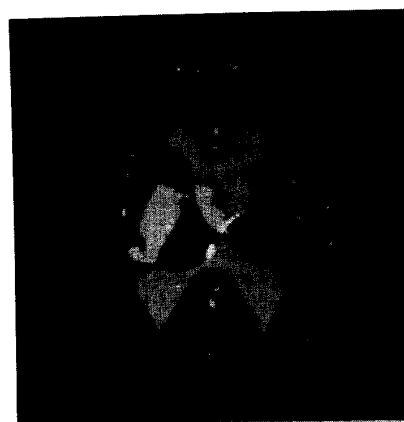


Figure 3: Combined segmentation map